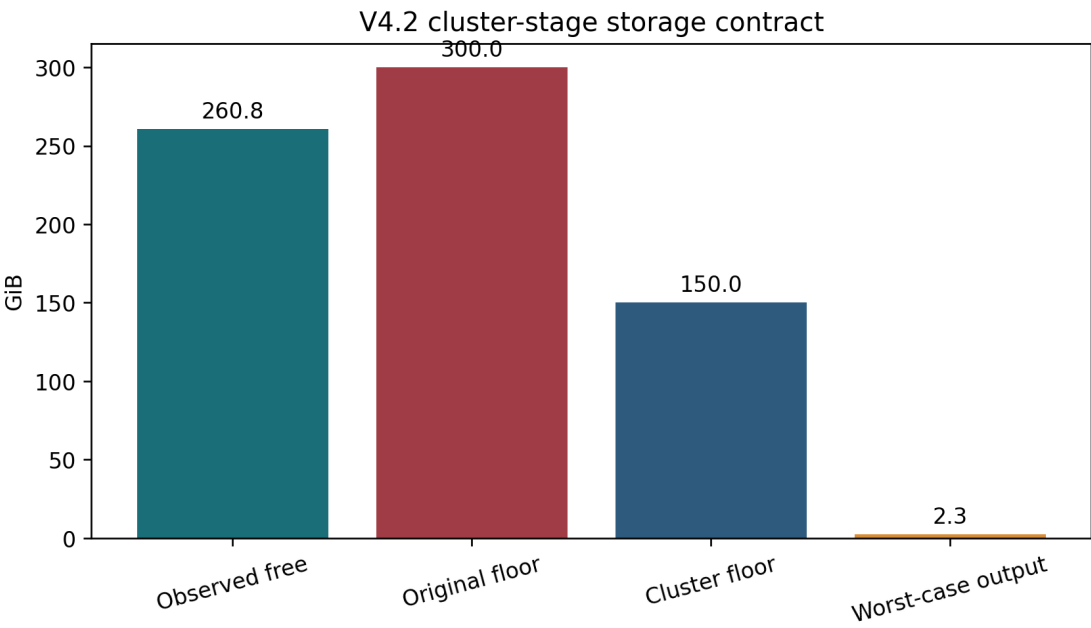


gdTAI V4.2 cluster-stage resource amendment

Decision

PASS_REVIEW_REQUIRED. Sparse staging and A100 scVI fitting completed under the original **300 GiB** SSD floor. RAPIDS clustering did not start because an unrelated shared-SSD BAM sort reduced free space. After that process completed and self-cleaned, free space stabilized at **260.8 GiB**.

The amendment retains 300 GiB for `prepare` and `fit`, and freezes **150 GiB** for `cluster` and `consensus`. This is an operational storage guard only. It does not change cells, HVGs, expression, scVI parameters, latent coordinates, clustering grids, random seeds, anchors, pseudo-label rules, or validation roles.



Storage contract

Completed evidence

- Staged matrix: **4,023,462 cells x 4,000 HVGs, 1,667,132,819** nonzero counts.
- Locked cohorts in staged object: **no**.
- Primary NK anchors: **21,054**.
- scVI latent: **4,023,462 x 30**, finite, A100, no CPU fallback.
- Source H5AD size/mtime checks: **6/6 unchanged**.
- Worst-case uncompressed partition payload plus reserve: **2.33 GiB**.
- Free-space margin above the amended floor and reserve: **108.4 GiB**.

Checks

check	status	detail
prepare_passed	PASS	/home/tanikai/databank/publicdata/tools/output_geo_tcell_research/Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_nk_reference/prepare_summary.json
fit_passed	PASS	/home/tanikai/databank/publicdata/tools/output_geo_tcell_research/Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_nk_reference/fit_summary.json
staged_h5ad_present	PASS	/ssd/tnk_phase3/Integrated_dataset/gdtai_v4_2_nk_reference/development_hvg_counts.h5ad
staged_h5ad_sha256	PASS	cbd6d60f919fd235d578fe5423a0b19134e7447ab36bec231dcc064976e0aa9d
staged_shape	PASS	(4023462, 4000)
staged_sparse_csr	PASS	csr_matrix
staged_nnz	PASS	1667132819 nnz
locked_cohorts_absent	PASS	prepare summary false
latent_present	PASS	/ssd/tnk_phase3/Integrated_dataset/gdtai_v4_2_nk_reference/X_scVI.npy
latent_sha256	PASS	7030b28be885b85a4b24ce2223bb16f7c7288a29195f2512bf7bbba330791ccf
latent_shape_and_finite	PASS	shape=(4023462, 30); finite=True
gpu_without_cpu_fallback	PASS	NVIDIA A100 80GB PCIe
scvi_model_present	PASS	/ssd/tnk_phase3/Integrated_dataset/gdtai_v4_2_nk_reference/scvi_model/model.pt
clustering_not_started	PASS	/ssd/tnk_phase3/Integrated_dataset/gdtai_v4_2_nk_reference/cluster_partitions.npz
source_h5ads_unchanged	PASS	6/6
stage_specific_floor_preserves_original_fit_floor	PASS	prepare/fit=300.0 GiB
cluster_floor_met	PASS	free=260.8; floor=150.0 GiB
cluster_output_reserve_met	PASS	post-reserve margin=108.4 GiB

Supervision boundary

Activating `CLUSTER_EXECUTION_APPROVAL.json` authorizes only the already approved RAPIDS repeated clustering and pseudo-NK consensus QC using the saved staged H5AD and latent representation. It does not authorize re-staging, re-fitting scVI, classifier fitting, threshold selection, promotion, release fitting, or atlas inference.

Artifacts

- Checks: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_2_cluster_resource_preflight/resource_amendment_checks.csv`
- Approval template: `Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_cluster_resource_preflight/CLUSTER_EXECUTION_APPROVAL_TEMPLATE.json`
- HTML: `gdT_prediction/gdtai_v4_2_cluster_resource_preflight/index.html`
- PDF: `gdT_prediction/gdtai_v4_2_cluster_resource_preflight/gdtai_v4_2_cluster_resource_preflight_report.pdf`